

Thermal degradation of wire — Solution

X25 (2025) — English translation

A1

0.20 pts

Set the current through source 1.4 A. Blow on the wire and choose the correct option from the answer sheets. Explain why this happened.

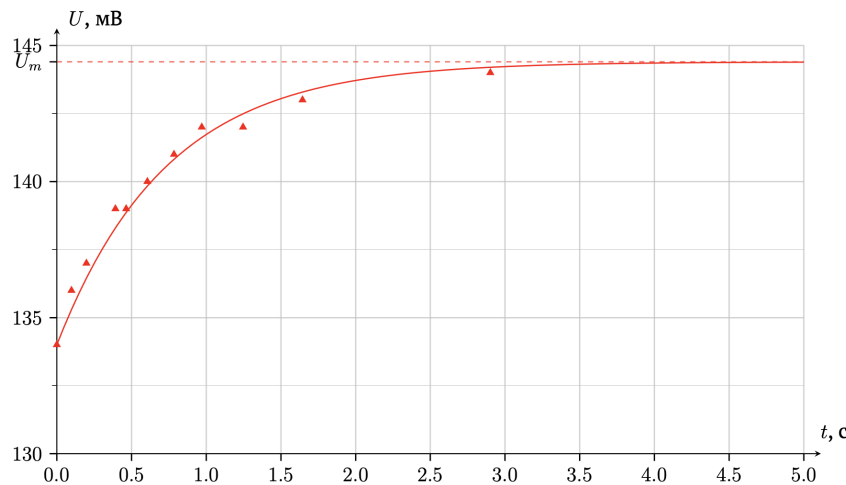
Answer: If you blow on the wire, the voltage on it will decrease. This is explained by the fact that the resistivity drops when the temperature decreases

A2

1.00 pts

Read the dependence of the voltage on the wire on time in the range 0-5 c after connecting to the source at a current strength of $I = 1.0$ A through it. Indicate the characteristic time during which the voltage on the wire is established.

t , ms	0.00	0.10	0.20	0.39	0.46	0.61	0.78	0.97	1.25	1.64	U_{avg} , mV	134	136	137	139	139	140	141	142
$U_m - U$																			
$\ln \frac{U_m - U}{1 \text{ mV}}$																			
	2.30	02.08	1.95	1.61	1.61	1.39	1.10	0.69	0.69	0.00									



A3

0.60 pts

Remove the dependence of the steady-state voltage U_m through the wire on the current strength $I < 1.5$ A through the wire. Take at least 15 measurements.

I , A	0.25	0.29	0.37	0.40	0.48	0.56	0.64	0.70	0.77	0.90	0.99	1.06	1.15	1.29	1.46	1.47	1.49	1.31	1.36
U_m , mV	32.1	37.5	47.2	51.7	62.4	73.3	84.7	92.6	102.7	123.3	137.7	150.3	166.8	193.1	228	233	236	196	207
$U_m I$, mW	8.0	10.9	17.5	20.7	30.0	41.0	54.2	64.8	79.1	111.0	136.3	159.3	191.8	249.1	332.9	342.5	351.6	256.8	281.5
U_m/I , mΩ	128.4	129.3	127.6	129.3	130.0	130.9	132.3	132.3	133.4	137.0	139.1	141.8	145.0	149.7	156.2	158.5	158.4	149.6	152.2

A4

0.20 pts

Theoretically, you will get the dependence $T(U)$. Express your answer using α, I, L, r_0, T, U .

The wire resistance at temperature T is equal to:

$$R(T) = \frac{\alpha T L}{\pi r_0^2} = \frac{U}{I}$$

Answer:

$$T(U) = \frac{\pi r_0^2}{\alpha I L} \cdot U$$

A5

0.30 pts

Write down the heat balance equation for the wire. Express your answer in terms of $U, I, c_m, L, r_0, \rho_m, \beta, T, T_0, \frac{dT}{dt}$.

$$U I dt = \rho_m \pi r_0^2 L c_m dT + \beta \cdot 2\pi r_0 L (T - T_0) dt$$

Answer:

$$U I = \rho_m \pi r_0^2 L c_m \frac{dT}{dt} + \beta \cdot 2\pi r_0 L (T - T_0)$$

A6

0.30 pts

Theoretically relate the steady-state voltage U_m and current I through the wire. Write the answer using $\alpha, T_0, r_0, L, \beta, U_m, I$.

When $U = U_m$: $T = T(U_m) = \text{const}$, it means:

$$U_m I = \beta \cdot 2\pi r_0 L (T(U_m) - T_0)$$

From point A4 we express $T(U_m)$ and substitute:

$$U_m I = \beta \cdot 2\pi r_0 L \left(\frac{\pi r_0^2}{\alpha L} \cdot \frac{U_m}{I} - T_0 \right)$$

$$U_m I = \beta \cdot 2\pi r_0 L \left(\frac{\pi r_0^2}{\alpha L} \cdot \frac{U_m}{I} - T_0 \right)$$

Answer:

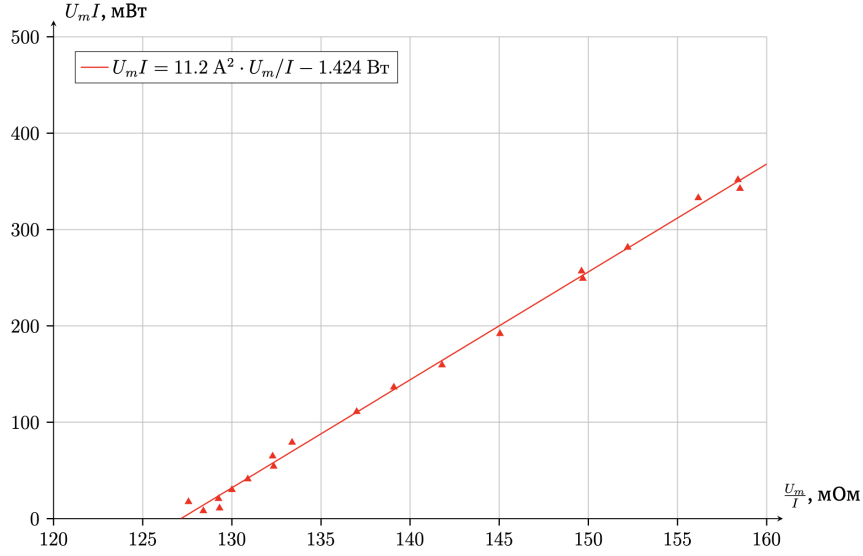
$$U_m I = \beta \cdot 2\pi r_0 L \left(\frac{\pi r_0^2}{\alpha L} \cdot \frac{U_m}{I} - T_0 \right)$$

A7

1.20 pts

Plot a graph of U_m and I in axes in which it is linear. From the graph, obtain the value of the heat transfer coefficient β .

Let's build a graph in axes $U_m I$ and $\frac{U_m}{I}, \beta = \frac{\alpha k_{\text{ang}}}{2\pi^2 r_0^3}$



A8

0.60 pts

Using the results of points A4 and A5, obtain the theoretical dependence of the voltage on the wire U on time t . Express your answer using $U_0, \alpha, \beta, I, r_0, L, T_0, \rho_m, c_m, U, t$.

$$\begin{aligned}
 UI &= \rho_m \pi r_0^2 L c_m \cdot \frac{\pi r_0^2}{\alpha I L} \cdot \frac{dU}{dt} + \beta \cdot 2\pi r_0 L \left(\frac{\pi r_0^2}{\alpha I L} \cdot U - T_0 \right) \\
 U \left(I - \frac{2\beta \pi^2 r_0^3}{\alpha I} \right) + 2\beta \pi r_0 L T_0 &= \rho_m c_m \cdot \frac{\pi^2 r_0^4}{\alpha I} \cdot \frac{dU}{dt} \\
 \int_0^t dt &= \rho_m c_m \cdot \frac{\pi^2 r_0^4}{\alpha I} \cdot \int_{U_0}^{U(t)} \frac{dU}{U \left(I - \frac{2\beta \pi^2 r_0^3}{\alpha I} \right) + 2\beta \pi r_0 L T_0} \\
 t &= \frac{\rho_m c_m \pi^2 r_0^4}{\alpha I^2 - 2\beta \pi^2 r_0^3} \cdot \ln \frac{U \left(I - \frac{2\beta \pi^2 r_0^3}{\alpha I} \right) + 2\beta \pi r_0 L T_0}{U_0 \left(I - \frac{2\beta \pi^2 r_0^3}{\alpha I} \right) + 2\beta \pi r_0 L T_0} \\
 U \left(I - \frac{2\beta \pi^2 r_0^3}{\alpha I} \right) + 2\beta \pi r_0 L T_0 &= \left(U_0 \left(I - \frac{2\beta \pi^2 r_0^3}{\alpha I} \right) + 2\beta \pi r_0 L T_0 \right) \cdot \exp \left(\frac{\alpha I^2 - 2\beta \pi^2 r_0^3}{\rho_m c_m \pi^2 r_0^4} \cdot t \right) \\
 U(t) &= \left(U_0 + \frac{2\beta \pi r_0 L T_0}{I - \frac{2\beta \pi^2 r_0^3}{\alpha I}} \right) \cdot \exp \left(\frac{\alpha I^2 - 2\beta \pi^2 r_0^3}{\rho_m c_m \pi^2 r_0^4} \cdot t \right) - \frac{2\beta \pi r_0 L T_0}{I - \frac{2\beta \pi^2 r_0^3}{\alpha I}} \\
 U(t) &= U_0 - \left(U_0 + \frac{2\beta \pi r_0 L T_0}{I - \frac{2\beta \pi^2 r_0^3}{\alpha I}} \right) \cdot \left(1 - \exp \left(\frac{\alpha I^2 - 2\beta \pi^2 r_0^3}{\rho_m c_m \pi^2 r_0^4} \cdot t \right) \right)
 \end{aligned}$$

Answer:

$$U(t) = U_0 + \left(\frac{2\alpha I \beta \pi r_0 L T_0}{2\beta \pi^2 r_0^3 - \alpha I^2} - U_0 \right) \cdot \left(1 - \exp \left(-\frac{2\beta \pi^2 r_0^3 - \alpha I^2}{\rho_m c_m \pi^2 r_0^4} \cdot t \right) \right)$$

A9

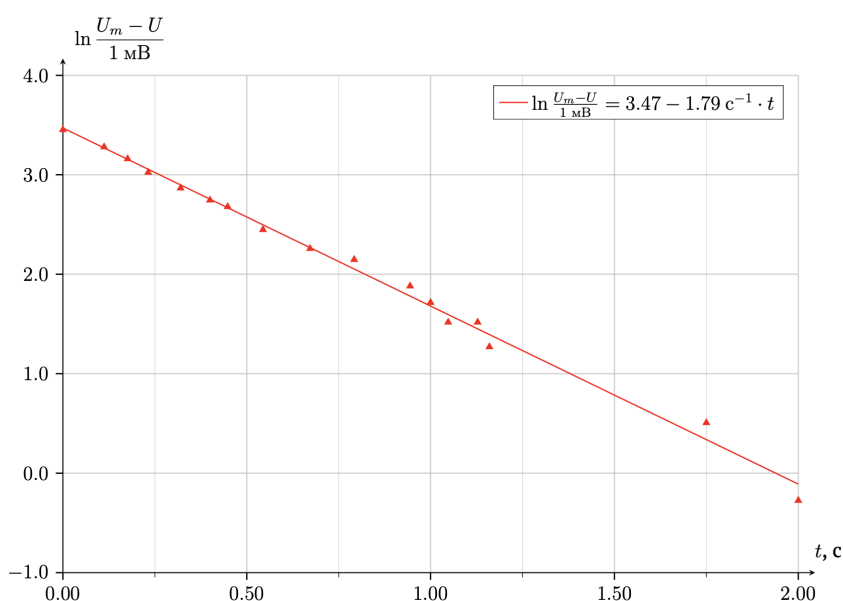
0.50 pts

Plot a linearized graph of U versus t measured in A2. From the graph, determine the specific heat capacity of copper c_m .

$$U(t) = U_0 + (U_m - U_0) \cdot \left(1 - \exp \left(-\frac{2\beta\pi^2 r_0^3 - \alpha I^2}{\rho_m c_m \pi^2 r_0^4} \cdot t \right) \right)$$

$$1 - \frac{U(t) - U_0}{U_m - U_0} = \exp \left(-\frac{2\beta\pi^2 r_0^3 - \alpha I^2}{\rho_m c_m \pi^2 r_0^4} \cdot t \right)$$

$$\ln \frac{U_m - U(t)}{U_m - U_0} = -\frac{2\beta\pi^2 r_0^3 - \alpha I^2}{\rho_m c_m \pi^2 r_0^4} \cdot t$$



A10

0.10 pts

The modern theory of heat capacity of solids states that the molar heat capacity of solids is approximately equal to $3R$. Estimate the specific heat capacity c_m^{th} from this theory.

$$c_m^{th} = \frac{3R}{\mu} = 392.6 \frac{\text{J}}{\text{kg} \cdot \text{K}}$$

B1

4.00 pts

Take measurements according to the algorithm above for 5 different values of I from the specified range.

$I = 2.00 \text{ A}$

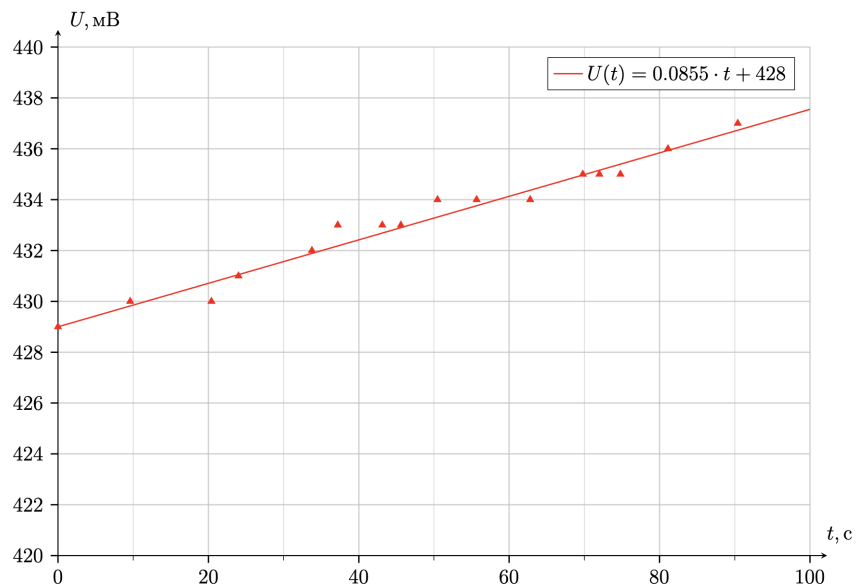
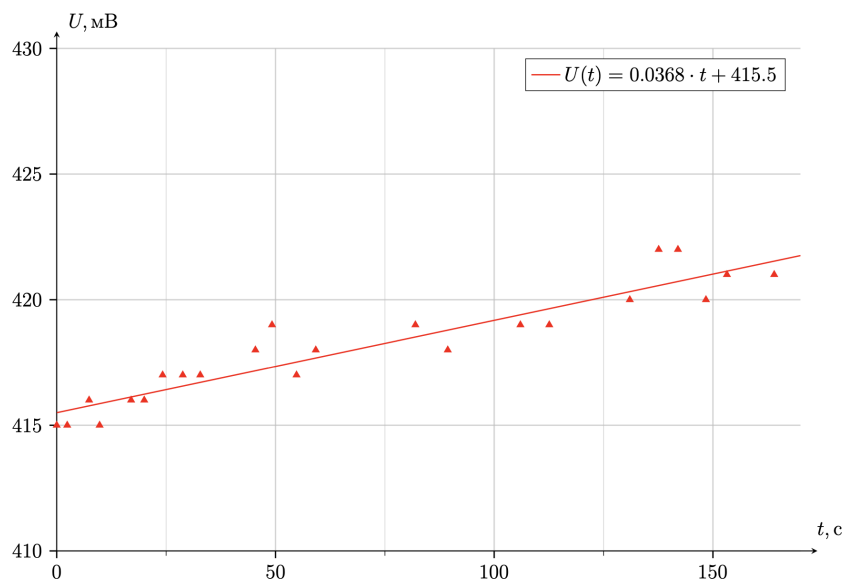
$I = 2.15 \text{ A}$

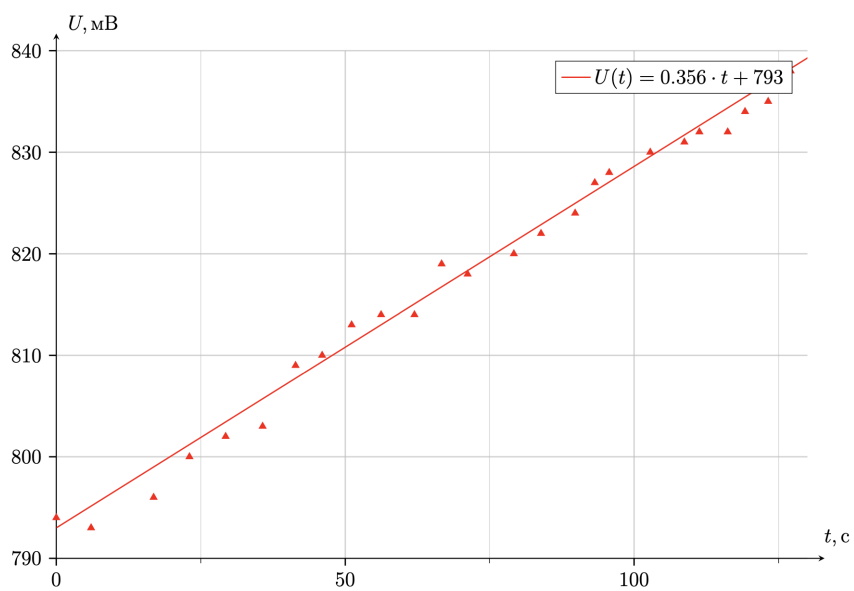
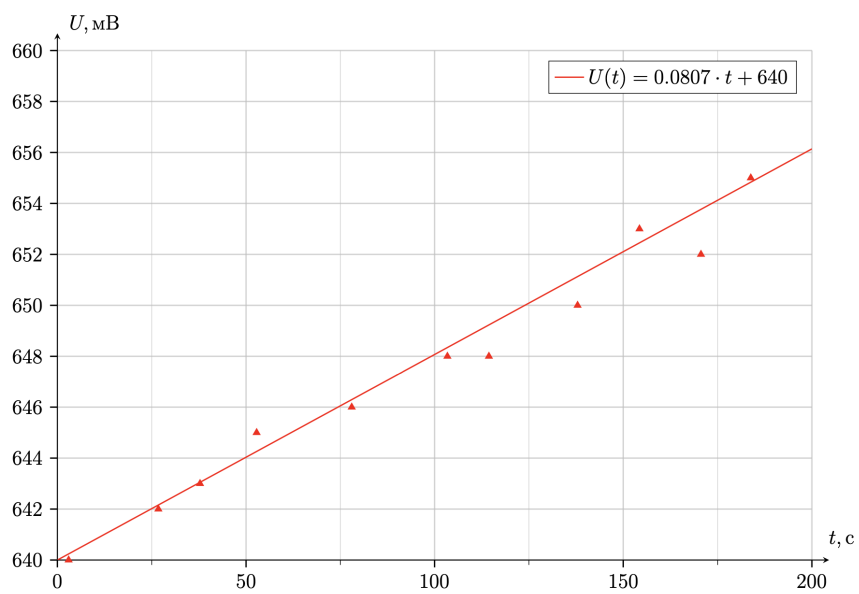
$I = 2.30 \text{ A}$

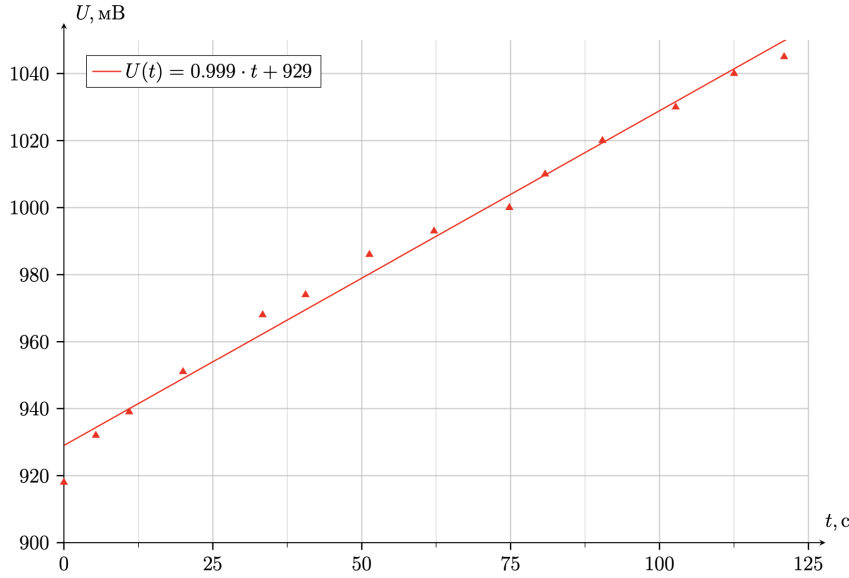
$I = 2.45 \text{ A}$

$I = 2.60 \text{ A}$

n	I, A	R_n, Ω	U_0, V	$\gamma(I), \text{mV/s}$	L, m	$\delta, \text{m/s}$	T_m, K	$\ln \delta$	$1/RT_m, \frac{\text{mol}}{\text{J}}$	1	2.00	0.105	0.416	0.037	0.0482
1	1.12	10^{-09}	458.9	-20.61	$2.62 \cdot 10^{-04}$	2	2.15	0.105	0.428	0.086	0.0482	$2.63 \cdot 10^{-09}$	502.3	-19.75	$2.40 \cdot 10^{-04}$
3	2.30	0.110	0.640	0.081	0.0505	$1.25 \cdot 10^{-09}$	559.0	-20.50	$2.15 \cdot 10^{-04}$	4	2.45	0.265	0.766	0.356	0.0405
5	10^{-09}	635.6	-18.44	$1.89 \cdot 10^{-04}$	5	2.60	0.136	0.919	0.999	0.0624	$1.05 \cdot 10^{-09}$	744.0	-18.38	$1.62 \cdot 10^{-04}$	







B2

0.10 pts

Express the length L_n of wire sample number n in terms of α, T_0, r_0, R_n . Using the data obtained earlier, recalculate L_n .

$$R_n = \frac{\alpha T_0 L_n}{\pi r_0^2}$$

Answer:

$$L_n = \frac{\pi r_0^2 R_n}{\alpha T_0}$$

B3

0.10 pts

Express the temperature of the wire T in terms of I, β, L, U, r_0, T_0 .

$$UI = 2\beta\pi r_0 L(T - T_0)$$

Answer:

$$T = T_0 + \frac{UI}{2\beta\pi r_0 L}$$

B4

0.10 pts

Express the voltage on the wire U in terms of $I, \alpha, L, T(U)$ and the radius of the conducting part of the wire $r(t)$.

Answer:

$$U = \frac{\alpha T(U) \cdot IL}{\pi r(t)^2}$$

B5

0.70 pts

Using the results of steps B3 and B4, obtain the theoretical relationship $r(t)$. Express your answer using $I, L, \alpha, \beta, T_0, r_0, t, \gamma(I), U_0$.

$$U_0 + \gamma(I)t = \frac{\alpha IL}{\pi r(t)^2} \cdot \left(T_0 + \frac{(U_0 + \gamma(I)t)I}{2\beta\pi r_0 L} \right)$$

Answer:

$$r(t) = \sqrt{\frac{\alpha IL}{\pi(U_0 + \gamma(I)t)} \cdot \left(T_0 + \frac{(U_0 + \gamma(I)t)I}{2\beta\pi r_0 L}\right)}$$

B6

0.50 pts

Expand the resulting $r(t)$ in a Taylor series in powers of t to a linear term: $r(t) = r_0 - \delta t$. Express δ in terms of $r_0, \gamma(I), \alpha, T_0, L, I, U_0$.

$$r^2(t) \approx \frac{\alpha IL}{\pi} \left(\frac{I}{2\beta\pi r_0 L} + \frac{T_0}{U_0 + \gamma t} \right) \approx \frac{\alpha IL}{\pi} \left(\frac{I}{2\beta\pi r_0 L} + \frac{T_0}{U_0} - \frac{\gamma T_0}{U_0^2} t \right)$$

$$r(t) = \sqrt{\frac{\alpha IL}{\pi} \left(\frac{I}{2\beta\pi r_0 L} + \frac{T_0}{U_0} \right) \cdot \left(1 - \frac{\gamma T_0 t}{\frac{IU_0^2}{2\beta\pi r_0 L} + U_0 T_0} \right)}$$

$$r(t) = \sqrt{\frac{\alpha IL}{\pi} \left(\frac{I}{2\beta\pi r_0 L} + \frac{T_0}{U_0} \right) \cdot \left(1 - \frac{\gamma T_0 t}{\frac{IU_0^2}{\beta\pi r_0 L} + 2U_0 T_0} \right)}$$

$$r(t) = r_0 \cdot \left(1 - \frac{\gamma T_0 t}{\frac{IU_0^2}{\beta\pi r_0 L} + 2U_0 T_0} \right)$$

$$\delta = \frac{\alpha IL r_0 \gamma T_0}{2U_0^2 \pi r_0^2}$$

Answer:

$$\delta = \frac{\alpha IL \gamma T_0}{2U_0^2 \pi r_0}$$

B8

0.20 pts

Specify dimension E_A .

Answer:

$$[E_A] = \frac{\text{J}}{\text{mol}}$$

B9

1.30 pts

Find E_A .

$$T_m = T(U_m) = \frac{\pi r_0^2}{\alpha IL} \cdot U_m = \frac{2\pi^2 r_0^3 \beta T_0}{-\alpha I^2 + 2\beta\pi^2 r_0^3}$$

$$\ln \delta = \ln \delta_0 - \frac{E_A}{RT_m}$$

Answer:

$$E_A = 22.9 \frac{\text{kJ}}{\text{mol}}$$

